YANG XU @ZJU CS

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EDUCATION

Zhejiang University Hangzhou, China
Undergraduate Student 2022 - 2026 *(expected)*

- **Major**: Computer Science and Technology.
- **Minor Program**: Advanced Honor Class for Engineering Education(only 50 science&engineering students selected in each Grade), Chu Kochen Honors College, Zhejiang University.
- **GPA**: 3.99/4.3

RESEARCH INTERESTS

My current research interests focus on designing novel model architectures, particularly efficient attention mechanisms, and on scaling large language models. Previously, I worked on interpretability and trustworthy machine learning, motivated by a desire to uncover the underlying "physics" of neural networks and to ensure their controllability and reliability. My research direction has shifted from DECONSTRUCTION to CONSTRUCTION, with the belief that "What I cannot create, I do not understand". The next paradigm in artificial intelligence will not emerge from merely refining existing frameworks, but from building fundamentally new ones. My long-term goal is to develop efficient, scalable models with genuine agency.

EXPERIENCES

MiniMax | Shanghai, China 2026.02-Now

- **Internship:** Algorithm & Engineering Co-design.
- **Focus:** Pretraining and Scaling of Large Language Models.

Summer Research Program | Beijing, China 2025.07-2025.08

- **Mentored by Jingyang Yuan** (Researcher at Deepseek, the best paper author of ACL 2025)
- **Focus:** Triton-based kernel optimization; Efficient attention mechanisms (linear and sparse attention)

UW-Madison | Madison, USA 2025.03-2025.11(remote)

- **Remote Research Intern** in the School of Computer, Data & Information Sciences
- **Advisor:** Prof. Sharon Li.
- **Focus:** Evaluation of LLM Trustworthiness, Collaboration with Amazon

Rutgers University | New Jersey, USA 2024.07-2025.02(remote since 2024.9)

- **Visiting Student and Research Intern** in the Department of Computer Science.
- **Advisor:** Prof. Hao Wang.
- **Focus:** Mechanistic Interpretability Study of LLMs.

Shanghai Jiao Tong University | Shanghai, China 2023.05-2024.09(remote)

- **Remote Research Intern** in the John Hopcroft Center for Computer Science, School of electronic information and electrical engineering.
- **Advisor:** Prof. Quanshi Zhang.
- **Focus:** Interpretability of Neural Networks and Deep Learning Theory.

Projects

Fused RoPE & Co-RoPE (Exploration)

- **Summary**: Triton Implementation of Fused RoPE and Exploration of Contextual Improvement of RoPE(Co-RoPE)
- **Link**: https://github.com/Superposition09m/RoPE-CoRoPE

Stanford CS336: Language Modeling from Scratch - A1

- **Summary**: A modular, from-scratch PyTorch implementation featuring modern LM architectural components: BPE tokenization, MHA, RMSNorm, SwiGLU activations, RoPE, and custom AdamW optimizer. Validated on TinyStories (loss 1.34) and OpenWebText (loss 3.78).
- **Link**: https://github.com/Superposition09m/CS336-Mine(course website: https://stanford-cs336.github.io/spring2025/)

Publications and Preprints

1. **Yang Xu***, Xuanming Zhang*, Samuel Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Sharon Li. *Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions*. **ICLR 2026.**
2. **Yang Xu**, Yi Wang, Hao Wang. *Tracking the Feature Dynamics in LLM Training: A Mechanistic Study*. arXiv preprint arXiv:2412.17626, 2024.
3. Qihan Ren*, Junpeng Zhang*, **Yang Xu**, Yue Xin, Dongrui Liu, Quanshi Zhang. *Towards the Dynamics of a DNN Learning Symbolic Interactions*. Neural Information Processing Systems (NeurIPS), 2024.
4. Xu Cheng*, Lei Cheng*, Zhaoran Peng, **Yang Xu**, Tian Han, Quanshi Zhang. *Layerwise Change of Knowledge in Neural Networks*. Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235:8038-8059, 2024.

Awards and Honors

- **2nd Scholarship** in Zhejiang University.
- **Win 1st Prize for twice(2022, 2023)** in Zhejiang Division of National Mathematics Competition for College Students.

English Proficiency

TOEFL iBT: 102 (Listening: 30, Reading: 28, Speaking: 20, Writing: 24) *March 2024*

Activities: Member of ZJUFLA (Zhejiang University Foreign Language Association), English Corner Organizer for 2 semesters.

Sept. 2023 – June 2024

Academic Services

Reviewer for: *International Conference on Learning Representations (ICLR) 2025, North American Chapter of the Association for Computational Linguistics (NAACL) 2025, ICLR 2026.*